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Department of Electrical and Electronics Engineering

MATLAB Laboratory Manual for Power Systems-I

Academic Year 2025-2026

LIST OF EXPERIMENTS:

Sl. No	Name of the Experiment	Hrs.
1.	Determination of transmission line parameters (inductance and capacitance) for a single-phase line using MATLAB Program.	3
2.	Determination of transmission line parameters (inductance and capacitance) for a symmetrical and unsymmetrical three-phase transmission line using MATLAB Program.	3
3.	Determination of Power Angle Diagrams, Reluctance Power, Excitation Emf and Regulation for salient and non-salient pole Synchronous machine using MATLAB Program.	3
4.	Modeling and performance analysis of short and medium transmission line using MATLAB Program.	3
5.	Study of surge impedance loading of a long transmission line using MATLAB Program.	3
6.	Verify the Ferranti effect in a transmission line using MATLAB Program.	3
7.	Evaluation of sag in overhead transmission lines with equal and unequal tower heights using MATLAB Program.	3
8.	Determination of string efficiency of suspension-type insulators using MATLAB Program.	3
9.	Improvement of voltage profile at a load bus using a shunt capacitor using SIMULINK Model.	3
10.	Study and compare the performance of ring and radial distributions systems using SIMULINK Model.	3

EXPERIMENT NO- 1

AIM: Determination of transmission line parameters (inductance and capacitance) for a single-phase line.

SOFTWARE REQUIRED: MATLAB

THEORY: Transmission line has four parameters – resistance, inductance, capacitance and conductance. The inductance and capacitance are due to the effect of magnetic and electric fields around the conductor. The resistance of the conductor is best determined from the manufacture data, the inductances and capacitances can be evaluated using the formula.

For a single conductor (two-wire system), the inductance per unit length is given by:

$$L = 2 \times 10^{-7} \ln \left(\frac{D}{r'} \right) \text{ H/m}$$

Where,

L = inductance per meter (H/m)

D = distance between the two conductors (m)

r' = GMR (Geometric Mean Radius) or equivalent radius, calculated as:

$$r' = 0.7788 \times r$$

r = Physical radius of the conductor

The capacitance per unit length of a single-phase line is:

$$C = \frac{2 \pi \epsilon_0}{\ln (D/r)} \text{ F/m}$$

Where:

C = capacitance per meter (F/m)

$\epsilon_0 = 8.854 \times 10^{-12}$ (permittivity of free space)

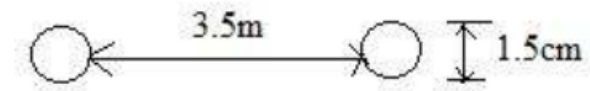
D = distance between the two conductors (m)

r = radius of the conductor (m)

PROCEDURE:

1. Enter the command window of the MATLAB.
2. Create a new M – file by selecting File - New – M – File
3. Type and save the program in the editor window.
4. Execute the program by either pressing Tools – Run.
5. View the results.

Question: Calculate the loop inductance and capacitance of a 1 phase line with two parallel conductors spaced 3.5m apart. The diameter of each conductor is 1.5 cm.



MATLAB PROGRAM FOR CALCULATION OF INDUCTANCE AND CAPACITANCE OF SINGLE-PHASE LINE

```
clc;
clear all;
disp('CALCULATION OF INDUCTANCE AND CAPACITANCE OF 1 PHASE LINE');
d=input('Enter diameter in cm:');
r=d/2;
rad=r*10^(-2);
D=input('Enter distance between conductors in m:');
r1=rad*0.7788;
L=4*10^(-7)*log(D/r1);
C=(pi*8.854*10^(-12))/(log(D/rad));
disp('INDUCTANCE(in H/m):');
disp(L);
disp('CAPACITANCE(in F/m):');
disp(C);
```

OUTPUT:

CALCULATION OF INDUCTANCE AND CAPACITANCE OF 1 PHASE LINE

Enter diameter in cm: 1.5

Enter distance between conductors in m: 3.5

INDUCTANCE (in H/m): 2.5582e-006

CAPACITANCE (in F/m): 4.5261e-012

EXPERIMENT NO- 2

AIM: Determination of transmission line parameters (inductance and capacitance) for a symmetrical and unsymmetrical three-phase transmission line using MATLAB Program.

SOFTWARE REQUIRED: MATLAB

THEORY: Transmission line has four parameters – resistance, inductance, capacitance and conductance. The inductance and capacitance are due to the effect of magnetic and electric fields around the conductor. The resistance of the conductor is best determined from the manufactures data, the inductances and capacitances can be evaluated using the formula.

- i. For a symmetrical three-phase transmission line, the inductance per unit length is given by:

$$L = 2 \times 10^{-7} \ln \left(\frac{D}{r'} \right) \text{ H/m}$$

Where,

L = inductance per meter (H/m)

D = Distance between any two conductors (since spacing is symmetrical)

r' = GMR (Geometric Mean Radius) or equivalent radius, calculated as:

$$r' = 0.7788 \times r$$

r: Physical radius of the conductor

The capacitance per unit length of a single-phase line is:

$$C = \frac{2 \pi \epsilon_0}{\ln (D/r)} \text{ F/m}$$

Where:

C = capacitance per meter (F/m)

$\epsilon_0 = 8.854 \times 10^{-12}$ (permittivity of free space)

D = Distance between conductors (same for all pairs in symmetrical case)

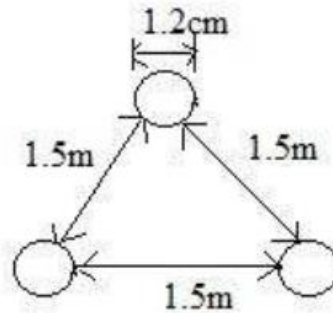
r = radius of the conductor (m)

PROCEDURE:

1. Enter the command window of the MATLAB.
2. Create a new M – file by selecting File - New – M – File
3. Type and save the program in the editor window.
4. Execute the program by either pressing Tools – Run.

5. View the results.

Question: Calculate the inductance and capacitance of a conductor of a 3 phase system shown which has 1.2 cm diameter and conductors at the edge of an equilateral triangle of side 1.5m.



MATLAB PROGRAM FOR CALCULATION OF INDUCTANCE AND CAPACITANCE OF THREE PHASE SYMMETRIC LINE

```
clc;
clear all;
disp('CALCULATION OF INDUCTANCE AND CAPACITANCE OF 3 PHASE
SYMMETRIC LINE');
d=input ('Enter diameter in cm:');
r=d/2;
rad=r*10^(-2);
D=input ('Enter distance between conductors in m:');
r1=rad*0.7788.
L=2*10^(-7) *log(D/r1); C=(2*pi*8.854*10^(-12))/(log(D/rad));
disp('INDUCTANCE (in H/m):');
disp(L);
disp('CAPACITANCE (in F/m):');
disp(C);
```

OUTPUT:

CALCULATION OF INDUCTANCE AND CAPACITANCE OF 3 PHASE SYMMETRIC
LINE

Enter diameter in cm: 1.2

Enter distance between conductors in m: 1.5

INDUCTANCE (in H/m): 1.1543e-006

CAPACITANCE (in F/m): 1.0075e-011

- ii. For an unsymmetrical transposed three-phase transmission line, the inductance per unit length is given by:

$$L = 2 \times 10^{-7} \ln \left(\frac{D_{eq}}{r'} \right) \text{ H/m}$$

L = inductance per meter (H/m)

$D_{eq} = (D_{ab} \cdot D_{bc} \cdot D_{ca})^{1/3}$: GMD (Geometric Mean Distance) between the conductors (m)

$r' = \text{GMR (Geometric Mean Radius) or equivalent radius, calculated as:}$

$r' = 0.7788 \times r$; GMR of the conductor

r = Physical radius of the conductor

D_{ab}, D_{bc}, D_{ca} Distance between the conductor

For an unsymmetrical transposed three-phase transmission line, the capacitance per unit length is given by:

$$C = \frac{2\pi\epsilon_0}{\ln \left(\frac{D_{eq}}{r} \right)} \text{ F/m}$$

C : Capacitance per phase (F/m)

$\epsilon_0 = 8.854 \times 10^{-12}$ (F/m): Permittivity of free space

$D_{eq} = (D_{ab} \cdot D_{bc} \cdot D_{ca})^{1/3}$: GMD

r = Radius of the conductor (m)

PROCEDURE:

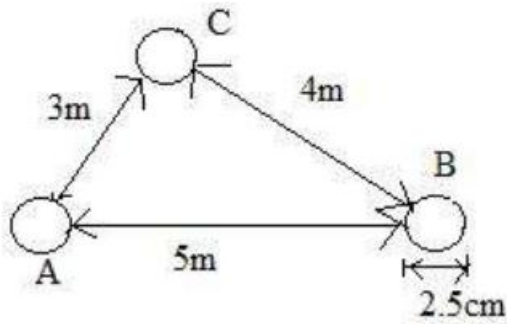
1. Enter the command window of the MATLAB.
2. Create a new M – file by selecting File - New – M – File
3. Type and save the program in the editor window.
4. Execute the program by either pressing Tools – Run.
5. View the results.

Question: Calculate the inductance, capacitance and reactance of 3 phase 50 Hz overhead transmission line which has conductors of 2.5cm diameter.

Distance between conductors are

5m between A & B 4m between B & C 3m between C & A

Assume conductors are transposed regularly



MATLAB PROGRAM FOR CALCULATION OF INDUCTANCE AND CAPACITANCE OF THREE PHASE UNSYMMETRIC TRANSPOSED LINE

```
clc;
```

```
clear all;
```

```
disp('CALCULATION OF INDUCTANCE AND CAPACITANCE OF 3 PHASE  
UNSYMMETRIC LINE - TRANSPOSED');
```

```
d=input('Enter diameter in cm:');
```

```
r=d/2;
```

```
rad=r*10^(-2);
```

```
Dab=input('Enter distance between conductors A & B in m:');
```

```
Dbc=input('Enter distance between conductors B & C in m:');
```

```
Dca=input('Enter distance between conductors C & A in m:');
```

```
f=input('Enter Frequency');
```

```
Deq=(Dab*Dbc*Dca)^(1/3); r1=rad*0.7788; L=2*10^(-7)*log(Deq/r1);
```

```
C=(2*pi*8.854*10^(-12))/(log(Deq/rad));
```

```
disp('INDUCTANCE (in H/m):');
```

```
disp(L);
```

```
disp('CAPACITANCE (in F/m):');
```

```
disp(C);
```

```
XL=2*pi*f*L;
```

```
XC=1/(2*pi*f*C);
```

```
disp('INDUCTIVER REACTANCE(in ohm/m):');
```

```
disp(XL);
```

```
disp('CAPACITIVE REACTANCE(in ohm/m):');
```

```
disp(XC);
```


OUTPUT:**CALCULATION OF INDUCTANCE AND CAPACITANCE OF 3 PHASE
UNSYMMETRIC LINE - TRANSPOSED**

Enter diameter in cm: 2.5

Enter distance between conductors A & B in m: 5

Enter distance between conductors B & C in m: 4

Enter distance between conductors C & A in m: 3

Enter Frequency50 INDUCTANCE (in H/m):1.1994e-006

CAPACITANCE (in F/m): 9.6804e-012

INDUCTIVER REACTANCE (in ohm/m): 3.7679e-004

CAPACITIVE REACTANCE (in ohm/m): 3.2882e+008

EXPERIMENT NO-3

AIM - Determination of the Power Angle Diagram, Reluctance Power, Excitation, EMF, and Regulation of Salient and Non-Salient Pole Synchronous Machines Using MATLAB.

APPARATUS/SOFTWARE REQUIRED:

1. MATLAB
2. Computer with basic programming capability,

THEORY:

A synchronous machine delivers power based on its internal excitation, terminal voltage, and load angle δ . The power output differs between salient pole (non-uniform air gap) and non-salient pole (uniform air gap) machines.

1. Non-Salient Pole Power Equation:

$$P = \frac{EV}{X_s} \sin \delta$$

2. Salient Pole Power Equation:

$$P = \frac{EV}{X_d} \sin \delta + \frac{V^2}{2} \left(\frac{1}{X_q} - \frac{1}{X_d} \right) \sin 2\delta$$

- The first term is electromagnetic power, and the second is reluctance power.

3. Voltage Regulation:

$$\text{Regulation} = \frac{E - V}{V} \times 100 \%$$

Where,

E = excitation EMF

V = terminal voltage

X_d, X_q = d-axis and q-axis reactances.

X_s = synchronous reactance (non-salient)

δ = power/load angle

PROCEDURE:

1. Open MATLAB and create a new script.
2. Copy and paste the following MATLAB code:

MATLAB Code:

% Experiment: Power Angle and Regulation Analysis for Synchronous Machines

% Parameters

```
V = 1;           % per unit terminal voltage
E = 1.2;         % excitation emf in per unit
X_d = 1.1;       % direct-axis reactance (salient)
X_q = 0.7;       % quadrature-axis reactance (salient)
X_s = 1.0;       % synchronous reactance (non-salient)
pf = 0.9;        % power factor (lagging)
P_load = 0.8;    % real power in per unit
```

% Power angle (delta) from 0° to 180°

```
delta_deg = linspace(0, 180, 1000);
delta_rad = deg2rad(delta_deg);
```

% Salient pole power components

```
P_em = (E * V / X_d) .* sin(delta_rad);
P_rel = (V^2 / 2) * (1/X_q - 1/X_d) * sin(2 * delta_rad);
P_total_sal = P_em + P_rel;
```

% Non-salient pole power

```
P_non = (E * V / X_s) .* sin(delta_rad);
```

%% Voltage Regulation Calculation

```
phi = acos(pf);           % Power factor angle
I_load = P_load / (V * pf); % Approximate per-unit current
```

% Regulation for non-salient

```
E_non_calc = sqrt( (V + I_load * X_s * sin(phi))^2 + (I_load * X_s * cos(phi))^2 );
reg_non = (E_non_calc - V) / V * 100;
```

% Regulation for salient

```
E_sal_calc = sqrt( (V + I_load * X_d * sin(phi))^2 + (I_load * X_q * cos(phi))^2 );
reg_sal = (E_sal_calc - V) / V * 100;
```

% Print results

```
fprintf('Voltage Regulation (Non-Salient): %.2f%%\n', reg_non);
fprintf('Voltage Regulation (Salient) : %.2f%%\n', reg_sal);
```

%% Plotting

figure;

```
plot(delta_deg, P_em, 'b--', 'LineWidth', 2); hold on;
plot(delta_deg, P_rel, 'm-.', 'LineWidth', 2);
plot(delta_deg, P_total_sal, 'r-', 'LineWidth', 2);
plot(delta_deg, P_non, 'k:', 'LineWidth', 2);
xlabel('Power Angle \delta (degrees)');
ylabel('Power Output (p.u.)');
title('Power-Angle Diagram: Salient vs Non-Salient');
legend('P_{em} (Salient)', 'P_{rel} (Salient)', 'P_{total} (Salient)', 'P_{non-salient}',
'Location', 'Best');
grid on;
xlim([0 180]);
ylim([-0.5 2]);
```

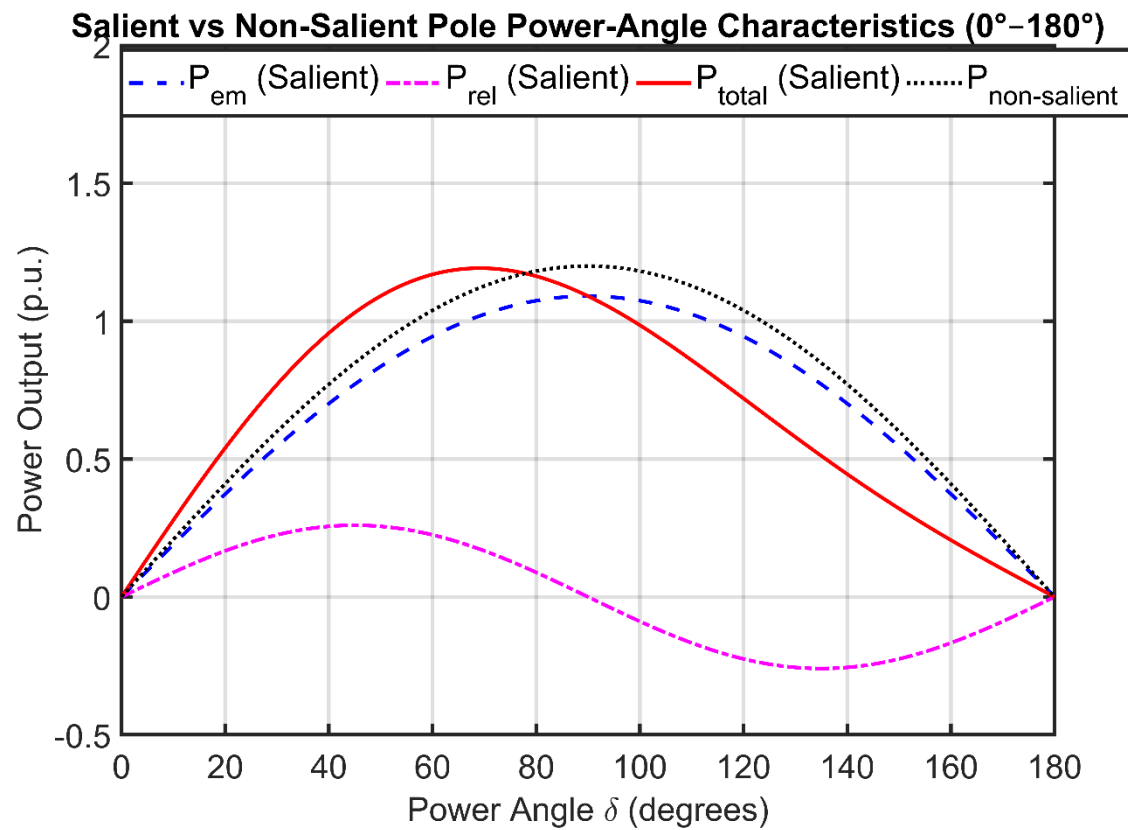
3. Run the script and observe:

- Power-angle curves
- Salient power components
- Printed voltage regulation for both machine types

RESULTS:

Type	Regulation (%)
Non-Salient Pole	20.12%
Salient Pole	23.85%

Power Curve :



CONCLUSION:

This experiment verified the theoretical expressions of synchronous machine power output. The salient pole machine exhibited a higher total power due to reluctance torque, while the non-salient displayed pure sinusoidal characteristics. Voltage regulation was higher in salient machines due to their magnetic saliency.

EXPERIMENT NO-4

AIM: Modeling and performance analysis of short transmission line

SOFTWARE REQUIRED: MATLAB

THEORY:

i. Short Transmission Line

FORMULA:

$$V_S = AV_R + BI_R$$

$$I_S = CV_R + DI_R$$

TYPE	METHOD	ABCD PARAMETERS
Short	-----	$A=D=1$; $B=Z$; $C=0$

PROCEDURE:

1. Enter the command window of the MATLAB.
2. Create a new M – file by selecting File - New – M – File
3. Type and save the program in the editor window.
4. Execute the program by either pressing Tools – Run.
5. View the results.

1. An overhead 3 phase transmission line delivers 4000KW at 11 KV at 0.8 pf lagging. The resistance and reactance of each conductor are 1.5Ω and 4Ω per phase. Determine the line performance.

Manual Calculation:

MATLAB PROGRAM:

```
clc;
clear all;
R=input('Resistance :');
XL=input('Inductive Reactance :');
XC=input('Capacitive Reactance :');
G=input('Conductance :');
length=input('Length of Transmission Line :');
f=input('Frequency :');
Zl = (R+j*XL)*length;
```

```

Y1=(G+j*XC)*length;
A = 1;
B = Z1;
C = 0;
D =1;
TM = [ A B; C D ];
VRL=input('ENTER RECEIVING END VOLTAGE :');
VRP=VRL/(sqrt(3));
PR = input('ENTER RECEIVING END LOAD IN MW :');
Pf=input('ENTER THE RECEIVING END LOAD POWER FACTOR :');
h=acos(Pf);
SR=PR/Pf;
SR=SR*(cos(h)+j*sin(h));
QR=imag(SR);
IR=conj(SR)/(3*conj(VRP));
SM=TM*[VRP;IR];
VS=SM(1,1);
IS=SM(2,1);
Pfs=cos(angle(VS)-angle(IS));
SS=3*VS*conj(IS);
VSA=angle(VS)*(180/pi);
ISA=angle(IS)*(180/pi);
VS=sqrt(3)*abs(VS);
IS=abs(IS)*1000;
VREG=((VS/(abs(TM(1,1)))-VRL)/VRL)*100;
PS=real(SS);
QS=imag(SS);
eff=PR/PS*100;
PL=PS-PR;
QL=QS-QR;
Z1
Y1
TM
fprintf('SENDING END LINE VOLTAGE %g at %g degrees \n',VS,VSA);

```

```

fprintf('SENDING END LINE CURRENT %g at %g degrees \n',IS,ISA);
fprintf('SENDING END POWER FACTOR %g\n',Pfs);
fprintf('SENDING END REAL POWER %g\n',PS);
fprintf('SENDING END REACTIVE POWER %g\n',QS);
fprintf('PERCENTAGE VOLTAGE REGULATION %g\n',VREG);
fprintf('REAL POWER LOSS %g\n',PL);
fprintf('REACTIVE POWER LOSS %g\n',QL);
fprintf('EFFICIENCY %G', eff);

```

OUTPUT:

Resistance : 1.5

Inductive Reactance : 4

Capacitive Reactance : 0

Conductance : 0

Length of Transmission Line : 1

Frequency : 50

ENTER RECEIVING END VOLTAGE : 11

ENTER RECEIVING END LOAD IN MW : 4

ENTER THE RECEIVING END LOAD POWER FACTOR : 0.8

$Z1 =$

$1.5000 + 4.0000i$

$Y1 =$

0

$TM =$

$1.0000 \ 1.5000 + 4.0000i$

0 1.0000

SENDING END LINE VOLTAGE 12.6795 at 4.72953 degrees

SENDING END LINE CURRENT 262.432 at -36.8699 degrees

SENDING END POWER FACTOR 0.747805

SENDING END REAL POWER 4.30992

SENDING END REACTIVE POWER 3.82645

PERCENTAGE VOLTAGE REGULATION 15.2685

REAL POWER LOSS 0.309917

REACTIVE POWER LOSS 0.826446

EFFICIENCY 92.8092

ii. MEDIUM TRANSMISSION LINE

FORMULA:

$$V_S = AV_R + BI_R$$

$$I_S = CI_R + DI_R$$

TYPE	METHOD	ABCD PARAMETERS
Medium	Nominal T Method	$A=D=1+YZ/2;$ $B=Z(1+YZ/4);$
	Nominal π Method	$C=Y; A=D=1+YZ/2; B=Z;$ $C=Y(1+YZ/4);$

2. A balanced 3 phase load of 30 MW is supplied at 132KV, 50Hz and 0.85 pf lag by means of a line. The series impedance is $20+j52\Omega$ and total admittance is $315 \times 10^{-6} \text{U}$. Using Normal T method determine A, B, C, D parameters and regulation.

Manual Calculation:

PROGRAM:

```
clc;
clear all;
R=input('Resistance :');
XL=input('Inductive Reactance :');
XC=input('Capacitive Reactance :');
G=input('Conductance :');
length=input('Length of Transmission Line :');
f=input('Frequency :');
Z1= (R+j*XL)*length;
Y1= (G+j*XC)*length;
m=menu('ENTER THE TYPE OF NETWORK','NOMINAL T', 'NOMINAL PI');
switch m
case {1}
A = 1+(Z1*Y1/2);
```

```

B=Z1*(1+(Z1*Y1/4));
C=Y1;
D=A;
otherwise
A = 1+(Z1*Y1/2);
B=Z1;
C=Y1*(1+(Z1*Y1/4));
D=A;
end
TM = [ A B; C D ];
VRL=input('ENTER RECEIVING END VOLTAGE :');
VRP=VRL/(sqrt(3));
PR = input('ENTER RECEIVING END LOAD IN MW :');
Pf=input('ENTER THE RECEIVING END LOAD POWER FACTOR :');
h=acos(Pf);
SR=PR/Pf;
SR=SR*(cos(h)+j*sin(h));
QR=imag(SR);
IR=conj(SR)/(3*conj(VRP));
SM=TM*[VRP;IR];
VS=SM(1,1);
IS=SM(2,1);
Pfs=cos(angle(VS)-angle(IS));
SS=3*VS*conj(IS);
VSA=angle(VS)*(180/pi);
ISA=angle(IS)*(180/pi);
VS=sqrt(3)*abs(VS);
IS=abs(IS)*1000;
VREG=((VS/(abs(TM(1,1)))-VRL)/VRL)*100;
PS=real(SS);
QS=imag(SS);
eff=PR/PS*100;
PL=PS-PR;
QL=QS-QR;

```

Z1

Y1

TM

fprintf('SENDING END LINE VOLTAGE %g at %g degrees \n',VS,VSA);

fprintf('SENDING END LINE CURRENT %g at %g degrees \n',IS,ISA);

fprintf('SENDING END POWER FACTOR %g\n',Pfs);

fprintf('SENDING END REAL POWER %g\n',PS);

fprintf('SENDING END REACTIVE POWER %g\n',QS);

fprintf('PERCENTAGE VOLTAGE REGULATION %g\n',VREG);

fprintf('REAL POWER LOSS %g\n',PL);

fprintf('REACTIVE POWER LOSS %g\n',QL);

fprintf('EFFICIENCY %G', eff);

NOMINAL (T)

OUTPUT:

Resistance : 20

Inductive Reactance : 52

Capacitive Reactance : 315×10^{-6}

Conductance : 0

Length of Transmission Line : 1

Frequency : 50

ENTER RECEIVING END VOLTAGE : 132

ENTER RECEIVING END LOAD IN MW : 30

ENTER THE RECEIVING END LOAD POWER FACTOR : 0.85

Z1 =

20.0000 +52.0000i

Y1 =

0 +3.1500e-004i

TM =

0.9918 + 0.0031i 19.8362 +51.8186i

0 + 0.0003i 0.9918 + 0.0031i

SENDING END LINE VOLTAGE 143.035 at 3.76761 degrees

SENDING END LINE CURRENT 142.007 at -23.3284 degrees

SENDING END POWER FACTOR 0.890244
SENDING END REAL POWER 31.3199
SENDING END REACTIVE POWER 16.0245
PERCENTAGE VOLTAGE REGULATION 9.25407
REAL POWER LOSS 1.31989
REACTIVE POWER LOSS -2.56785
EFFICIENCY 95.7858

Example: A 50Hz, 3 phase, 100 km transmission line has total impedance of 35Ω , reactance of 140Ω and shunt admittance of $930 \times 10^{-6} \text{ S}$. It delivers 40 MW at 220KV, 0.9 pf lag. Using nominal π determine A,B ,C,D Vs, VSA,ISA , pf, Ps Qs, η .

Manual Calculation:

NOMINAL PI (π)

OUTPUT:

Resistance : 20

Inductive Reactance : 52

Capacitive Reactance : 315×10^{-6}

Conductance : 0

Length of Transmission Line : 1

Frequency : 50

ENTER RECEIVING END VOLTAGE : 132

ENTER RECEIVING END LOAD IN MW : 30

ENTER THE RECEIVING END LOAD POWER FACTOR : 0.85

$Z1 =$

$20.0000 + 52.0000i$

$Y1 =$

$0 + 3.1500e-004i$

$TM =$

$0.9918 + 0.0031i$ $20.0000 + 52.0000i$

$-0.0000 + 0.0003i$ $0.9918 + 0.0031i$

SENDING END LINE VOLTAGE 143.099 at 3.77321 degrees

SENDING END LINE CURRENT 142.011 at -23.3709 degrees

SENDING END POWER FACTOR 0.889862

SENDING END REAL POWER 31.3214

SENDING END REACTIVE POWER 16.0584

PERCENTAGE VOLTAGE REGULATION 9.30284

REAL POWER LOSS 1.32135

REACTIVE POWER LOSS -2.53394

EFFICIENCY 95.7813

EXPERIMENT NO-5

AIM: Study of surge impedance loading of a long transmission line using MATLAB Program.

SOFTWARE REQUIRED: MATLAB

THEORY:

1. Surge Impedance (Z_c) is the characteristic impedance of a transmission line and is given by:

$$Z_c = \frac{V_{oc}}{I_{sc}}$$

where:

- Voc: Open-circuit voltage at the receiving end
 - Isc: Short-circuit current at the receiving end
2. **Surge Impedance Loading (SIL)** is the power that a transmission line can carry at unity power factor when the load impedance equals the surge impedance:

$$SIL = \frac{V_{ph}^2}{Z_c}$$

For a single-phase system, V_{ph} is the RMS phase voltage.

3. A **π -section model** of a transmission line consists of a series impedance $Z=R+j\omega L$ $Z=R + j\omega L$ and two shunt admittances $Y/2=j\omega C/2$ $Y/2 = j\omega C/2$ at the sending and receiving ends.

PROCEDURE:

1. Open MATLAB and create a new script file.
2. Define the line parameters: resistance R, inductance L, capacitance C, and source voltage V_s .
3. Compute the total series impedance Z and shunt admittance Y.
4. Simulate **open-circuit** condition:
5. Receiving end is disconnected.
6. Use voltage divider rule to calculate Voc.
7. Simulate **short-circuit** condition:
8. Receiving end is shorted.
9. Calculate current $I_{sc}=V_s/Z$

10. Calculate:

$$Z_c = V_{oc} / I_{sc},$$

$$SIL = V_s^2 / Z_c$$

MATLAB CODE:

% SURGE IMPEDANCE AND SIL CALCULATION - SINGLE PHASE PI SECTION

clc;

clear;

%% LINE PARAMETERS

```
f = 50; % Frequency in Hz
omega = 2*pi*f; % Angular frequency (rad/s)
L = 1e-3; % Inductance in H
C = 1e-6; % Capacitance in F
R = 1; % Resistance in Ohms
Z = R + 1i*omega*L; % Series impedance
Y = 1i*omega*C; % Total shunt admittance
Vs = 230; % Source voltage (RMS)
```

%% OPEN CIRCUIT

```
Ysh = Y / 2;
```

```
Zsh = 1 / Ysh;
```

```
Voc = Vs * Z / (Z + Zsh); % Open-circuit receiving-end voltage
```

%% SHORT CIRCUIT

```
Isc = Vs / Z; % Short-circuit current at receiving end
```

%% SURGE IMPEDANCE

```
Zc = abs(Voc / Isc); % Magnitude of surge impedance
```

%% SIL CALCULATION

```
SIL = (Vs^2) / Zc; % Surge Impedance Loading (Single-phase)
```

%% DISPLAY

```
fprintf('--- RESULTS ---\n');
```

```
fprintf('Open Circuit Voltage (Voc) = %.2f V\n', abs(Voc));
```

```
fprintf('Short Circuit Current (Isc) = %.2f A\n', abs(Isc));
```

```
fprintf('Surge Impedance (Zc) = %.2f Ohms\n', Zc);
```

```
fprintf('Surge Impedance Loading (SIL) = %.2f W\n', SIL);
```

RESULT:

Parameter	Value
Frequency (f)	50 Hz
Inductance (L)	1 mH
Capacitance (C)	1 μ F
Resistance (R)	1 Ω
Source Voltage (Vs)	230 V
Open Circuit Voltage	~228.85 V
Short Circuit Current	~3.26 A
Surge Impedance (Zc)	~70.15 Ω
SIL (Single-Phase)	~7549.75 W

CONCLUSION:

This experiment successfully demonstrates how the surge impedance and surge impedance loading can be evaluated using MATLAB for a single-phase transmission line modeled with a π -section. The results are consistent with theoretical expectations.

EXPERIMENT NO- 6

AIM: Observation of the ferranti effect in medium and long transmission lines.

SOFTWARE REQUIRED: MATLAB

THEORY: The **Ferranti effect** is the phenomenon where the receiving end voltage becomes **higher** than the sending end voltage in an **open-circuited or lightly loaded** transmission line, especially noticeable in **medium and long lines**.

This is primarily due to:

- Line **capacitance** storing charge and
- **Charging current** causing voltage rise across the inductance

More pronounced in:

- Long lines (> 250 km)
- Light or no load condition

Transmission Line Modeling:

Transmission lines can be modeled as:

- **Short lines (<80 km):** Lumped parameters (no Ferranti effect)
- **Medium lines (80–250 km): π -model**
- **Long lines (>250 km):** Distributed parameters (or **cascade of π -sections**)

For this experiment, we'll use the **π -model** and vary the length.

MATLAB PROGRAM:

```
clc;
clear;
% Parameters
Vs = 220e3;          % Sending end voltage (V)
f = 50;              % Frequency (Hz)
L_per_km = 1e-3;     % Inductance per km (H/km)
C_per_km = 0.01e-6;  % Capacitance per km (F/km)
lengths = [50 150 300]; % Transmission line lengths (km)
% Result storage
Vr_magnitudes = zeros(size(lengths));
fprintf('Length (km)\tReceiving Voltage (kV)\n');
fprintf('-----\n');
```

```

for i = 1:length(lengths)
    len = lengths(i);
    % Total L and C
    L = L_per_km * len;
    C = C_per_km * len;
    % ABCD parameters for  $\pi$ -model
    Z = 1j*2*pi*f*L;
    Y = 1j*2*pi*f*C;
    A = 1 + (Z*Y)/2;
    B = Z;
    C_ = Y * (1 + (Z*Y)/4);
    D = A;
    % Open-circuited ( $I_r = 0$ )
    Vr = Vs / A;
    Vr_magnitudes(i) = abs(Vr)/1000; % Convert to kV
    fprintf('%10d\t%18.2f\n', len, Vr_magnitudes(i));
end
% Plotting
figure;
plot(lengths, Vr_magnitudes, '-o', 'LineWidth', 2);
xlabel('Transmission Line Length (km)');
ylabel('Receiving End Voltage (kV)');
title('Ferranti Effect: Receiving Voltage vs Line Length');
grid on;

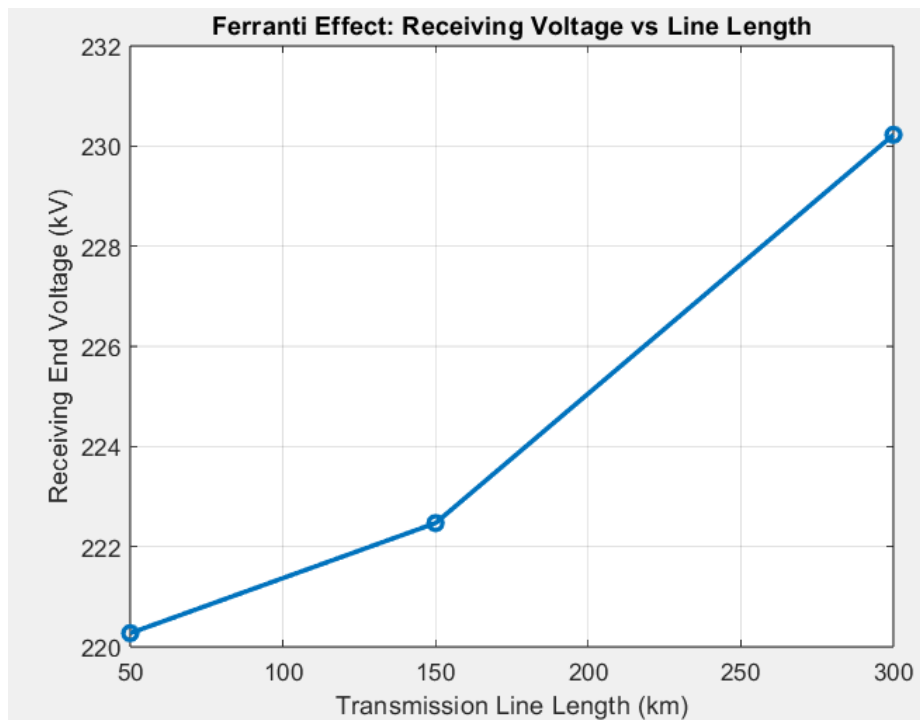
```

RESULT:

Line Length (km)	Receiving End Voltage (kV)
50	~220
150	>220
300	>>220

A line of 300 km may show a voltage rise of **10–15%** or more, depending on parameters.

Graphical representation:



CONCLUSION:

- The **Ferranti Effect** was clearly observed in medium and long lines.
- As the line length increases, the **receiving voltage increases** even without a load.
- The effect is caused by **line capacitance**, which stores charge and raises voltage under light/no-load conditions.
- In practical systems, this can damage equipment, hence **shunt reactors** or **voltage regulators** are used to mitigate the effect.

EXPERIMENT NO- 7

AIM: Determination of sag in overhead transmission lines with equal and unequal tower heights using MATLAB program.

SOFTWARE REQUIRED: MATLAB

1. Overhead Transmission Lines with Equal Tower Heights:

THEORY: Sag is the vertical distance between the highest point of the support (tower) and the lowest point of the conductor. In overhead transmission lines, sag is essential for maintaining proper clearance from the ground.

Assumptions:

- Towers are of equal height.
- The conductor is uniform and hangs under its own weight.
- No wind or ice loading (only self-weight considered).

Parabolic Approximation:

For small sag, the shape of the conductor can be approximated as a **parabola**.

$$\text{Sag (y)} = \frac{w L^2}{8T}$$

Where:

- w = weight per unit length of the conductor (N/m)
- L = span length between two towers (m)
- T = horizontal tension in the conductor (N)

2. Determination of sag in overhead transmission lines with unequal tower heights

In overhead transmission lines, **sag** refers to the vertical distance between the highest point of the support (tower) and the lowest point of the conductor. When tower heights are **unequal**, the lowest point of the conductor **shifts** from the center. The conductor still forms a **parabola or catenary**, depending on tension and span.

For unequal tower heights:

Let:

- h = difference in tower heights
- L = total horizontal span
- a = horizontal distance from lower tower to lowest point
- b = horizontal distance from higher tower to lowest point
- T = horizontal tension

- w = weight per unit length of conductor

Then:

$$a = \frac{L}{2} + \frac{hT}{wL}, \quad b = L - a$$

$$\text{Sag on left (from lower tower)} = \frac{w a^2}{2T}$$

$$\text{Sag on right (from higher tower)} = \frac{w b^2}{2T}$$

MATLAB CODE:

- Overhead Transmission Lines with Equal Tower Heights:

PROGRAM :

```
clc;
clear;
% Input parameters
L = 300;           % Span length in meters
w = 0.9;           % Weight per unit length of conductor (N/m)
T = 2500;          % Tension in the conductor (N)
% Calculate sag using parabolic approximation
sag = (w * L^2) / (8 * T);
% Display result
fprintf('Sag in the overhead line = %.2f meters\n', sag);
% Plotting the conductor profile
x = linspace(-L/2, L/2, 1000); % horizontal positions
y = (w .* x.^2) / (2 * T);      % parabolic profile
figure;
plot(x, y, 'b', 'LineWidth', 2);
xlabel('Horizontal Distance (m)');
ylabel('Vertical Sag (m)');
title('Sag Profile of Overhead Transmission Line');
grid on;
axis equal;
```

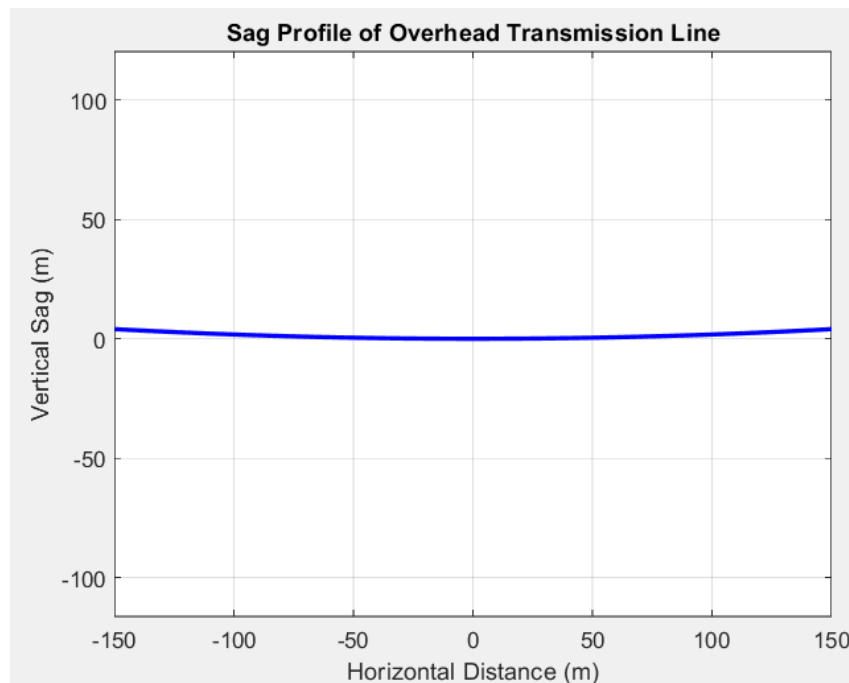
Result:

For:

- Span $L=300$ m
- Weight $w=0.9$ N/m
- Tension $T=2500$ N

$$\text{Sag} = \frac{w L^2}{8T}$$
$$= \frac{0.9 \times 300^2}{8 \times 2500} = 4.05\text{m}$$

So, the **sag is 4.05 meters**, and the plotted curve shows a symmetric parabolic conductor profile between the towers.



iii. Determination of sag in overhead transmission lines with unequal tower heights

PROGRAM:

% MATLAB script for sag determination in overhead transmission lines with unequal tower heights

clc;

clear;

% Input Parameters

$L = 300$; % Span length in meters

$h = 10$; % Height difference in meters (positive if right tower is higher)

$w = 1.2$; % Weight of conductor per unit length (N/m)

```

T = 2000;      % Horizontal tension in conductor (N)
% Calculating position of lowest point
a = (L/2) + (h*T)/(w*L);
b = L - a;
% Calculating sags
sag_left = (w * a^2) / (2 * T);
sag_right = (w * b^2) / (2 * T);
% Displaying results
fprintf('=== Sag Calculation for Unequal Tower Heights ===\n');
fprintf('Tower Height Difference (h): %.2f m\n', h);
fprintf('Span (L): %.2f m\n', L);
fprintf('Distance from Lower Tower to Lowest Point (a): %.2f m\n', a);
fprintf('Distance from Higher Tower to Lowest Point (b): %.2f m\n', b);
fprintf('Sag from Lower Tower: %.2f m\n', sag_left);
fprintf('Sag from Higher Tower: %.2f m\n', sag_right);

% Plotting the conductor profile
x = linspace(0, L, 1000);
y = zeros(size(x));
for i = 1:length(x)
    if x(i) <= a
        y(i) = (w/(2*T)) * (x(i) - a)^2 + sag_left;
    else
        y(i) = (w/(2*T)) * (x(i) - a)^2 + sag_left;
    end
end
plot(x, y, 'b-', 'LineWidth', 2);
title('Sag Profile for Unequal Tower Heights');
xlabel('Horizontal Distance (m)');
ylabel('Vertical Sag (m)');
grid on;
legend('Conductor Profile');
RESULT:
=== Sag Calculation for Unequal Tower Heights ===

```

Tower Height Difference (h): 10.00 m

Span (L): 300.00 m

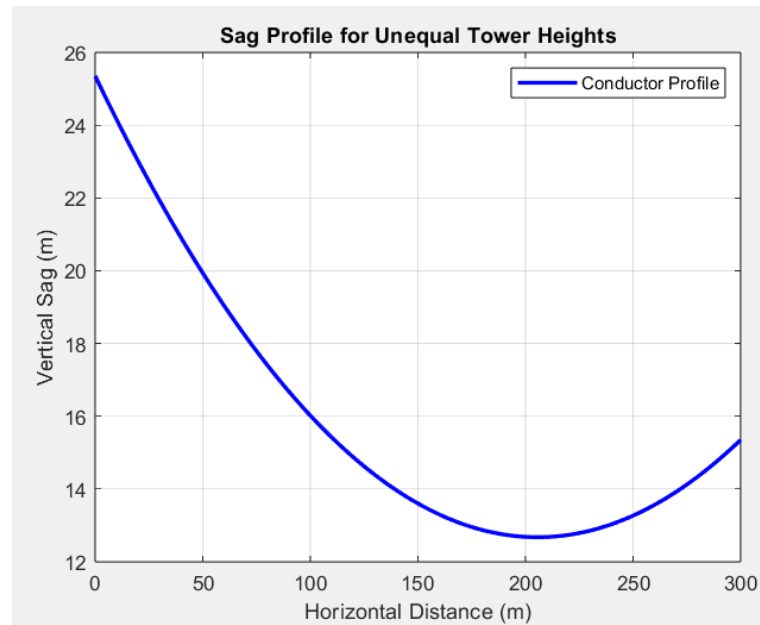
Distance from Lower Tower to Lowest Point (a): 205.56 m

Distance from Higher Tower to Lowest Point (b): 94.44 m

Sag from Lower Tower: 12.68 m

Sag from Higher Tower: 2.68 m

Graphical representation:



CONCLUSION

The sag in overhead transmission lines is directly influenced by the weight of the conductor and the span length, and inversely by the tension applied. In practical applications, factors such as safety clearances, thermal expansion, and wind loads must also be considered. The MATLAB experiment effectively demonstrates the relationship between sag, span, and tension, and visualizes the characteristic parabolic shape of the conductor. Maintaining appropriate sag is essential to ensure system reliability and prevent ground contact, particularly under varying environmental conditions.

EXPERIMENT NO-8

AIM: Calculation of string efficiency of suspension-type insulators using MATLAB program.

SOFTWARE REQUIRED: MATLAB

THEORY:

In high-voltage transmission systems, **suspension-type insulators** are used to support conductors. A set of insulator discs are connected in series to form a string.

Due to **non-uniform voltage distribution** (caused by stray capacitance between metal parts and tower/earth), the insulator closest to the conductor experiences the highest voltage. This leads to **reduced efficiency** of the string.

Definitions:

- **String Efficiency (η):**

$$\text{String Efficiency} = \frac{\text{Total voltage across string}}{n \times \text{Voltage across lowest unit}} \times 100\%$$

- **n:** Number of insulators in the string.
- **k:** Ratio of shunt capacitance (to earth or crossarm) to mutual capacitance between insulator units.

PROGRAM:

```
clc;
clear;
% Number of insulator discs
n = 5;
% Shunt capacitance ratio (k = shunt / mutual capacitance)
k = 0.2;
% Initialize voltage array
V = zeros(1, n);
% The system of equations is solved using matrix approach
A = zeros(n);
% Form the coefficient matrix based on voltage distribution
for i = 1:n
    for j = 1:n
        if i == j
            A(i,j) = 1 + (i-1)*k;
        elseif j == i+1
```

```

        A(i,j) = -1;
    end
end
end
% RHS of equations (last disc voltage assumed as 1 unit)
B = zeros(n,1);
B(n) = 1;
% Solve voltage ratios
volt_ratio = A\B;
% Normalize to total voltage
total_V = sum(volt_ratio);
V = volt_ratio / total_V;
% Calculate string efficiency
string_efficiency = (total_V / (n * V(n))) * 100;
% Display results
disp('Voltage across each disc (per unit of total voltage):');
disp(V);
disp(['Total String Efficiency: ', num2str(string_efficiency), ' %']);
% Plot
figure;
bar(1:n, V*100, 'b');
xlabel('Disc Number (from crossarm to conductor)');
ylabel('Voltage % of total');
title('Voltage Distribution Across Insulator String');
grid on;

```

RESULT:

For:

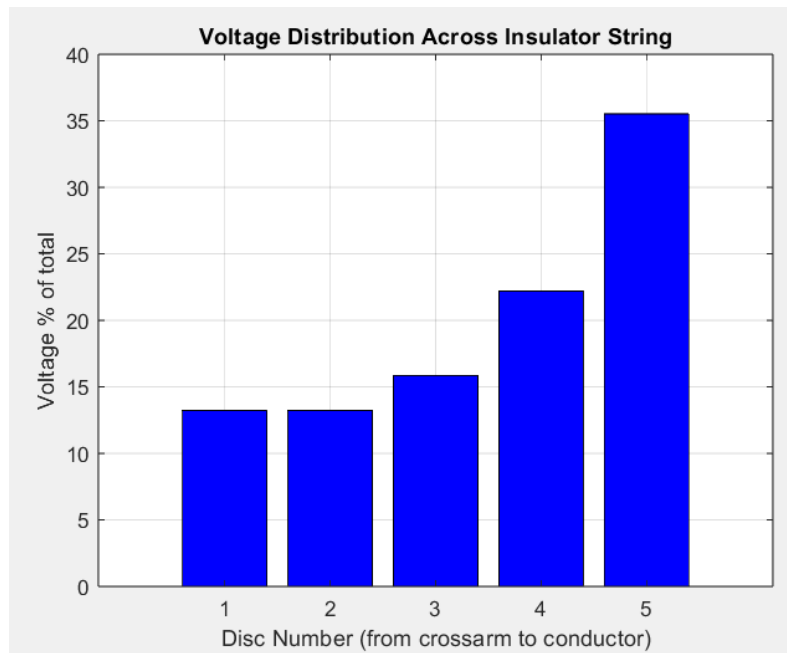
- $n = 5$
- $k = 0.2$

Voltage across each disc (per unit of total voltage):

0.1321 0.1321 0.1586 0.2220 0.3552

Total String Efficiency: 88.0767 %

Graphical Representation:



CONCLUSION:

The MATLAB simulation confirms that the **voltage is not uniformly distributed** across the discs in a suspension insulator string. The bottommost disc (nearest to the conductor) experiences **maximum voltage stress**, and the efficiency of the string is **below 100%** due to stray capacitances.

By using **guard rings or grading rings**, or reducing k , the voltage distribution can be made more uniform, thereby increasing string efficiency.

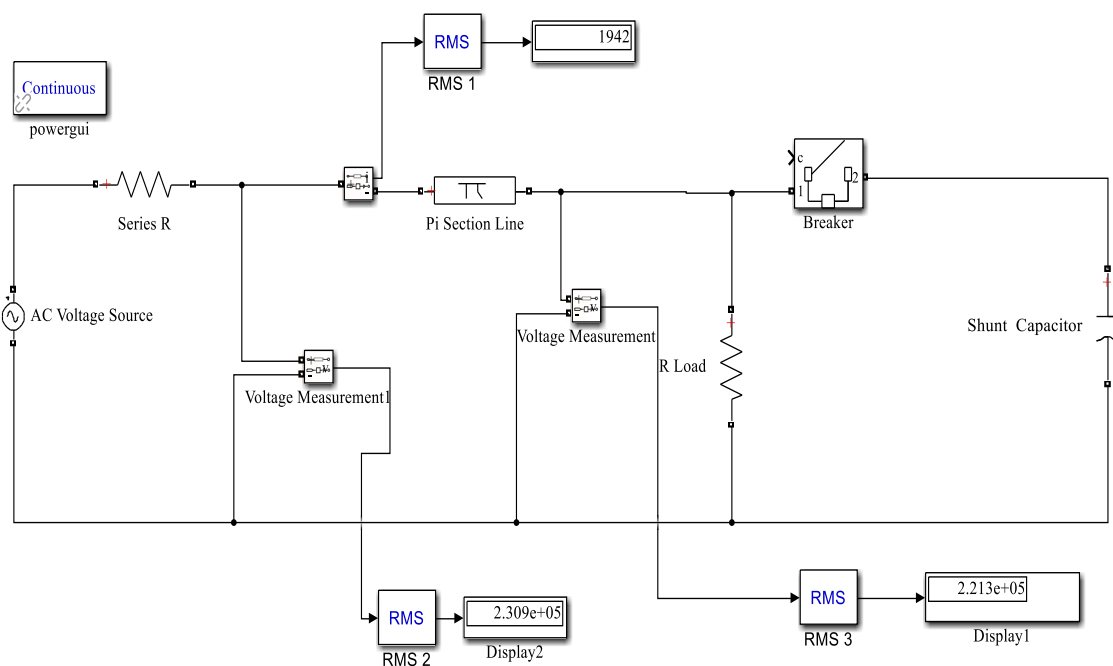
EXPERIMENT NO-9

AIM: - Improvement of voltage profile at a load bus using a shunt capacitor using MATLAB program.

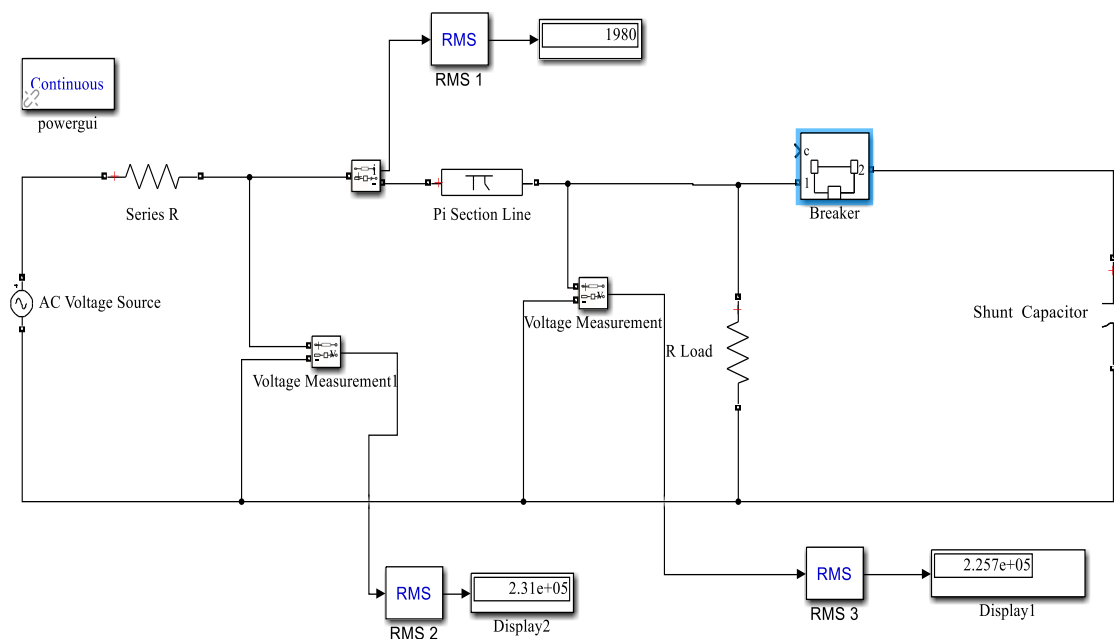
SIMULINK BLOCKS REQUIRED:

1. AC voltage sources in single phase, 50Hz
2. Pi section transmission line ($R = 0.01273 \text{ Ohms/km}$, $L = 0.9337 \text{ mH/km}$, $C = 12.74 \text{ nF/km}$, total length = 100km, 1 section).
3. R element representing a load ($R = 114 \text{ Ohms}$) - 1 No.
4. Element for a shunt capacitor - 1 No. ($C = 22.223 \mu\text{F}$)
5. Single phase breaker - 1 No
6. Breaker control block - 1 No
7. Voltage and current measurement blocks
8. RMS calculation blocks
9. Display block
10. Power Gui block

With Circuit Breaker Open (Without Shunt Capacitor):



With Circuit Breaker Closed (With Shunt Capacitor):



1. Build a SIMULINK model as shown in Circuit diagram.
2. Set the value of the ac voltage source to 326.598kV peak & note down the input and
3. Output voltage by keeping the circuit breaker switch to open (initial state 0).
4. Note down the input and Output voltage by keeping the circuit breaker switch to closed (initial state 1). That is the shunt capacitor is connected.
5. Note down change in output voltage.
6. Repeat same for other value of voltages.
7. Calculate the Voltage regulation of line with Capacitor and without capacitor.

OBSERVATION TABLE:

SL. No.	Vs	Without Capacitor		With Capacitor	
		I_L	V_R	I_L	V_R

RESULTS:

Voltage profile of transmission line with and without shunt capacitor has been plotted.

CONCLUSION:

For the above experiment, we observe the effect of radial distribution system.

EXPERIMENT NO-10

AIM- To study and compare the performance of a ring and radial distribution systems using SIMULINK model.

SIMULINK BLOCKS REQUIRED:

1. Singlephase325.27Vpeak,50HzACpowersupply
2. RL element representing distribution lines ($Z_1=Z_3=0.5+j0.26$ Ohms, $Z_2=0.25+j0.13$ Ohms)
3. R element representing loads($R=50$ Ohms)
4. Voltage and Current measurement blocks
5. RMS block
6. Display block
7. Powergui

CIRCUIT DIAGRAM:

Model of 500m long feeder is divided into three sections of 200m,100m, and 200m. Feeder is designed for a maximum load of 12A.

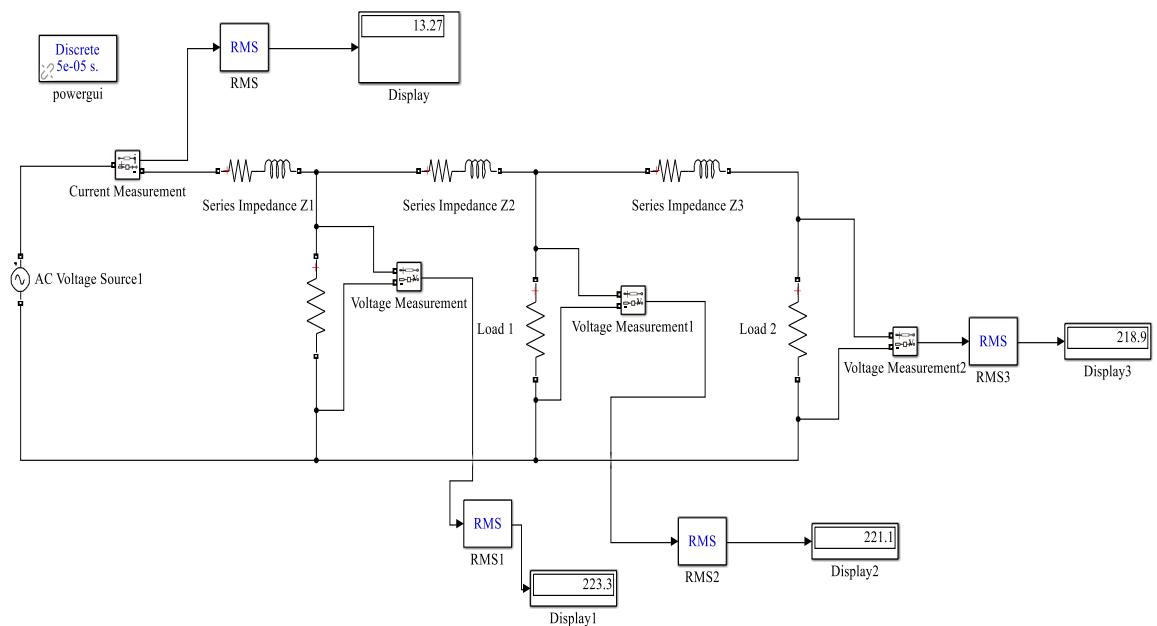


Fig.1 Radial Distribution system

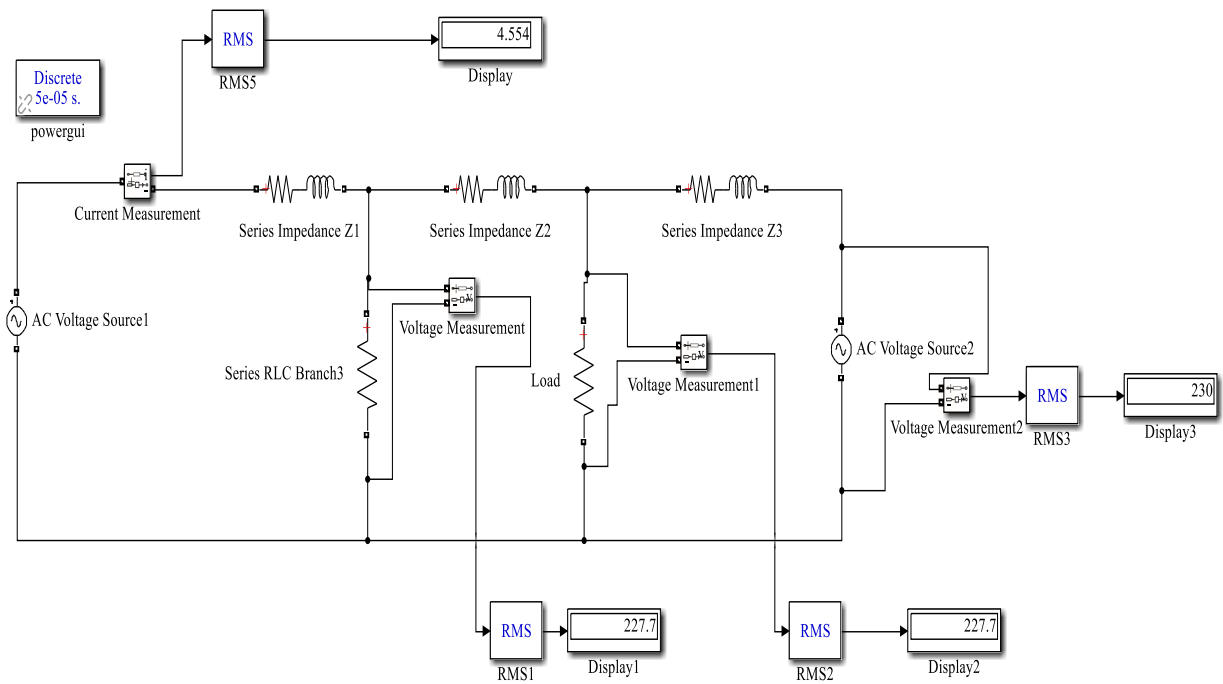


Fig.2 Ring Distribution system

PROCEDURE:

Radial System: -

1. Build the SIMULINK model as shown in the figure1.
2. Make sure that in the RMS block, the fundamental frequency is set to 50 HZ, initial
3. RMS value is set to 0, and sampling time is same as that used in the 'powergui' block.
4. Run the simulation model. Note down the reading so the ammeter (input side) and the three voltages.
5. The point a, b and c are the point after Z1, Z2 and Z3 impedance end point.

Ring Mains System: -

1. Build the SIMULINK model as shown in the figure2.
2. Make sure that in the RMS block, the fundamental frequency is set to 50HZ, initial
3. RMS value is set to 0, and sampling time is same as that used in the 'powergui' block.
4. Run the simulation model.
5. Note down the readings of the ammeter and the three voltages.
6. The point a, b are the point after Z1 and Z2 impedance end point.

OBSERVATION TABLE:

For Radial AC distribution System

Sl. No.	Vs (V)	Is (A)	Va (V)	% Voltage Drop	Vb (V)	% Voltage Drop	Vc (V)	% Voltage Drop

For ring type AC distribution system:

Sl. No.	Vs (V)	Is (A)	Va (V)	Vb (V)	% Voltage Drop

RESULT:**CONCLUSION:**

For the above experiment, we observe the effect of ring-main distribution system.

